Energy, Work & Power

ANSWER KEY

Section 1: Work Done by Constant & Inclined Forces

1 A force of 60 N pulls a box across a horizontal floor through a distance of 8 m. The force acts at an angle of 60° above the horizontal. Find the work done by the force.

a. 240 J ✓

b. 480 J

c. 415.7 J

d. 120 J

Work done $W = Fd\cos\theta = 60 \times 8 \times \cos(60^\circ) = 480 \times 0.5 = 240$ J.

2 A crate of mass 15 kg is lifted vertically at a constant speed through a height of 4 m. Taking $g = 10$ m/s^2, what is the work done against gravity?

a. 600 J ✓

b. 150 J

c. 300 J

d. 60 J

$W = mgh = 15(10)(4) = 600$ J.

3 A box is pushed 5 m along a horizontal floor against a constant frictional resistance of 35 N. What is the work done against friction?

a. 175 J ✓

b. 150 J

c. 350 J

d. 70 J

$W = F_f \times d = 35 \times 5 = 175$ J.

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Section 2: Kinetic Energy & Potential Energy

4 Calculate the kinetic energy of an athlete of mass 70 kg running at a speed of 8 m/s.

a. 2240 J ✓

b. 4480 J

c. 560 J

d. 280 J

$E_k = \frac{1}{2}mv^2 = \frac{1}{2}(70)(64) = 35 \times 64 = 2240$ J.

5 A rollercoaster car of mass 500 kg starts from rest at the top of a frictionless hill of height 20 m. Taking $g = 10$ m/s^2, find its speed at the bottom of the hill.

a. 20 m/s ✓

b. 10 m/s

c. 40 m/s

d. 14.1 m/s

Conservation of energy: $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2(10)(20)} = \sqrt{400} = 20$ m/s.

6 A bullet of mass 0.02 kg traveling at 300 m/s penetrates a wooden block to a depth of 0.15 m before stopping. What is the average resistive force exerted by the wood?

a. 6000 N ✓

b. 3000 N

c. 9000 N

d. 12000 N

Work-energy principle: $Fd = \frac{1}{2}mv^2 \Rightarrow F(0.15) = \frac{1}{2}(0.02)(90000) = 900 \Rightarrow F = 900/0.15 = 6000$ N.

Section 3: The Work-Energy Principle

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7 A skier of mass 60 kg starts from rest at the top of a slope 30 m high. When reaching the bottom, her speed is 20 m/s. Taking $g = 10\ \text{m/s}^2$, how much energy was lost due to friction?

a. 6000 J ✓

b. 12000 J

c. 18000 J

d. 3000 J

Initial potential energy $E_p = mgh = 60(10)(30) = 18000$ J. Final kinetic energy $E_k = \frac{1}{2}(60)(400) = 12000$ J. Energy lost $= 18000 - 12000 = 6000$ J.

8 A package of mass 4 kg slides down a rough incline of length 10 m and vertical drop 6 m. If the resistive force is a constant 8 N, find the speed of the package at the bottom (starting from rest, $g = 10\ \text{m/s}^2$).

a. $\sqrt{80} \approx 8.94$ m/s ✓

b. 10 m/s

c. 8 m/s

d. 12 m/s

Work done by gravity $= mgh = 4(10)(6) = 240$ J. Work against friction $= 8 \times 10 = 80$ J. Net kinetic energy $= 240 - 80 = 160$ J. $\frac{1}{2}(4)v^2 = 160 \Rightarrow 2v^2 = 160 \Rightarrow v^2 = 80 \Rightarrow v = \sqrt{80} \approx 8.94$ m/s.

9 A particle is projected up a rough incline (30°) with initial speed 10 m/s. If $\mu = 0.2$ and $g = 10\ \text{m/s}^2$, how far up the plane does the particle travel before coming to rest?

a. ≈ 7.43 m ✓

b. 10.0 m

c. 5.50 m

d. 8.25 m

Deceleration up slope: $a = g(\sin 30^\circ + \mu\cos 30^\circ) = 10(0.5 + 0.2 \times 0.866) = 10(0.5 + 0.1732) = 6.732\ \text{m/s}^2$. Distance $s = u^2/(2a) = 100/(2 \times 6.732) \approx 7.43$ m.

Section 4: Power and Vehicles in Motion

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10 A car of mass 1200 kg travels along a level road at a constant speed of 30 m/s. The engine is working at 45 kW. What is the total resistance to motion?

a. 1500 N ✓

b. 1200 N

c. 1350 N

d. 2000 N

$P = Fv \Rightarrow 45000 = F(30) \Rightarrow F = 1500$ N. Since speed is constant, driving force equals resistance: $R = 1500$ N.

11 The same car (mass 1200 kg, engine working at 45 kW) faces the same resistance of 1500 N while traveling at 20 m/s. What is its acceleration at this instant?

a. 0.625 m/s^2 ✓

b. 0.75 m/s^2

c. 1.00 m/s^2

d. 0.50 m/s^2

Driving force at 20 m/s: $F = P/v = 45000/20 = 2250$ N. Net force $= 2250 - 1500 = 750$ N. Acceleration $a = 750/1200 = 0.625$ m/s^2.

12 A truck of mass 5000 kg ascends an incline of 1 in 20 ($\sin\alpha = 1/20$) at a steady speed of 15 m/s. The engine power is 60 kW. Taking $g = 10$ m/s^2, find the resistance to motion.

a. 1500 N ✓

b. 2500 N

c. 4000 N

d. 2000 N

Driving force $F = P/v = 60000/15 = 4000$ N. Gravity component along slope $= mg\sin\alpha = 5000(10)(1/20) = 2500$ N. At steady speed: $F = mg\sin\alpha + R \Rightarrow 4000 = 2500 + R \Rightarrow R = 1500$ N.

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